NEUROSCIENCE

- It is the study of the nervous system which acts as a physical bridge between sensory input and motory output.

Anatomy & Function of a Neuron

Dendrites – the sensory part that receives signals from other neurons or the environment.

Soma (body) – Contains the nucleus and acts as an integration center.

Axon – the "cable" that transmits electric signals over distance.

Myelin – Electrical Insulation that allows signals to travel faster.

Axon Terminals – the motor part that releases chemical signals to the next cell.

Physics of Neural Signaling

- Membrane Potential : Neurons maintain a negative internal charge ($\sim 65 mV$) relative to the outside using ion pumps (Na/K pump)

- Action Potential: When a neuron is depolarized past a certain threshold, voltage-gated sodium channels open, causing a rapid spike in voltage.
Refractory Period: After firing, a neuron is temporarily hyper-polarized, making it difficult to fire again.
- Synapses: Communication between neurons happens at the synapse.

Electrical Synapses: Direct ion flow through gap junctions.
Chemical Synapses: Most common, an action potential triggers the release of neurotransmitters.
EPSP vs IPSP: Excitatory Signals (EPSP) push the neurons towards firing and Inhibitory Signals (IPSP) prevent it from firing.

Sensory & Motor Systems

Visual System: Early experiments showed that neurons in the primary visual cortex act as edge detectors, responding only to bars of light at specific orientations.

Receptive Fields: the specific area of space where a stimulus will trigger a response from a neuron

Auditory System: Neurons are tuned to specific frequencies of sound.
Motor System: Neurons in the motor cortex are tuned to specific directions of movement.

Network Motifs

- the brain uses recurring "microcircuits" to process information.

Lateral Inhibition - Active neurons silence their neighbors to sharpen a response. (crucial for vision)
Feedback loops - Can be used to stop signals (negative feedback) or create rhythmic oscillations like those seen during sleep (recurrent excitation)